Determinants of Household Participation in Securities Markets

An Empirical Analysis of the 2025 SEBI Household Survey

Shivansh Verma Sashwat Dhanuka

April 2026

Dept. of Computer Science, Ashoka University

The Problem

Only 25.5% of Indian households invest in securities.

Despite the post-2020 retail boom and zero-commission platforms, participation remains sharply stratified by socioeconomic position, geography, and information access.

Four research questions:

1. Who is likely to participate?
2. What keeps the 74.5% out?
3. Which securities do investors prefer, and for how long?
4. Can we predict these outcomes from basic profiles?

Sample

Total respondents	109,430
Active investors	27,882 (25.5%)
Non-investors	81,548 (74.5%)
Geographic zones	4
States / UTs covered	35

Methods

Logistic regression (HC3 robust SE, survey-weighted), K-means clustering, Histogram-based Gradient Boosting

Dataset and Schema

Source: 2025 SEBI Household Survey, nationally representative, pre-computed survey weights.

ETL pipeline: 449 raw columns transformed into boolean, integer, and ordinal types; loaded into a normalized PostgreSQL schema (17 domain-specific tables anchored by `respondent_id`).

Conditional routing:

- Investors: sub-module SS_B3 (info sources, motivations), $n = 9{,}797$
- Non-investors: sub-module SS_BB2 (barriers), $n = 14{,}794$

All group percentages computed against the routing-eligible population.

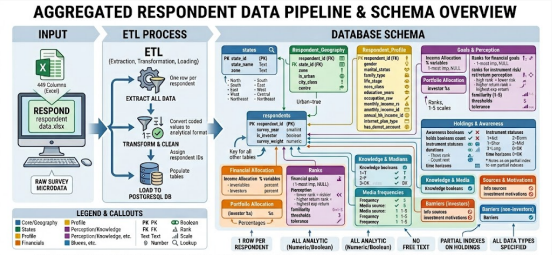

Figure 1: *

Schema architecture: modular, indexed, one-to-one with core respondent table.

EDA: Who Participates?

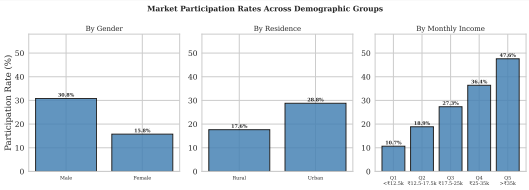

Figure 2: *

Participation rates by gender, residence, and monthly income quintile.

Steepest univariate signals:

- Income gradient: 10.7% (Q1, <Rs. 12,500/mo) to 47.6% (Q5, >Rs. 35,000/mo): a **4.5**× spread
- Gender gap: 30.8% (male) vs 15.8% (female), persists across all income levels
- Urban premium: 28.8% vs 17.6% (rural)
- Millennials (28–43) lead cohorts at 28.1%, exceeding both Gen Z (24.3%) and Gen X (22.9%)

These univariate signals motivate the multivariate regression: which factors

EDA: Awareness Gap and Holdings

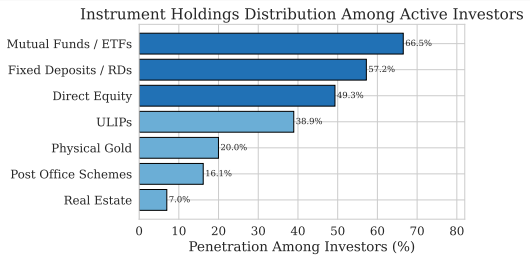

Figure 3: *

Instrument holdings among active investors ($n = 27{,}882$).

Awareness gradient (all 109,430 respondents):

- Fixed Deposits: **99.1%**
- Mutual Funds / ETFs: 67.0%
- Direct Equity: 62.0%
- Derivatives: **17.9%**

Holdings among investors:

- MF/ETF: 66.5% FD/RD: 57.2%
- Direct equity: 49.3%

MFs and FDs *co-dominate*: Indian retail investors layer market-linked products on top of traditional savings rather than replacing them.

Finding 1: What Drives Market Entry?

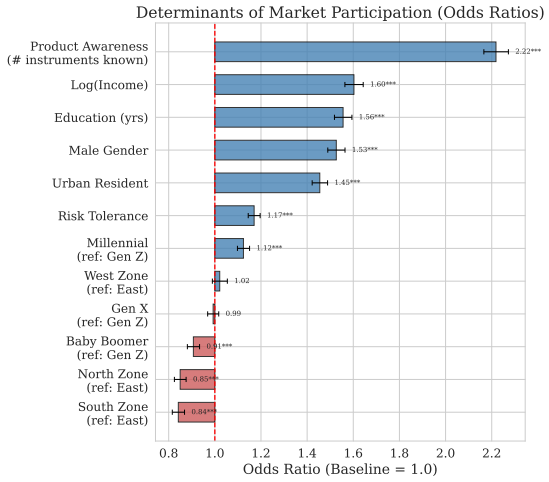

Figure 4: *

Product awareness is the single strongest predictor (OR = 2.22), exceeding:

Income (log)	OR = 1.60
Education	OR = 1.56
Male gender	OR = 1.53
Urban	OR = 1.45

Partial mediation: income and education partly work *through* awareness, but awareness retains a large independent effect after SES controls.

Two predictors return null results (Gen X, West zone): the model does not produce spurious significance at $N \approx 100{,}000$.

Policy implication

Finding 2: Three Archetypes of Non-Participation

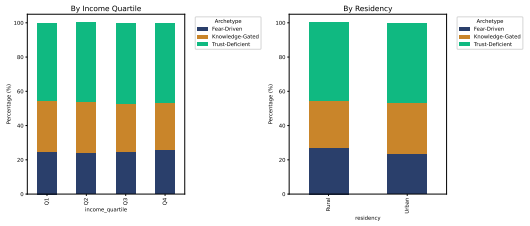

Figure 5: *

Demographic distribution of barrier archetypes ($n = 14{,}705$ non-investor barrier responses).

K-means on 18 self-reported barriers identifies three profiles:

Trust-Deficient (46.3%)

Persistent across income levels and urban cohorts. Requires institutional accountability: grievance redressal, IPF promotion, broker transparency.

Knowledge-Gated (29.0%)

Concentrated in rural areas and lowest income quartile. Needs geographically targeted vernacular literacy delivery.

Fear-Driven (24.8%)

Responds to loss-framing education and first-loss protection onramps (SGB, RBI

Finding 3: Behavioral Traps Inside the Market

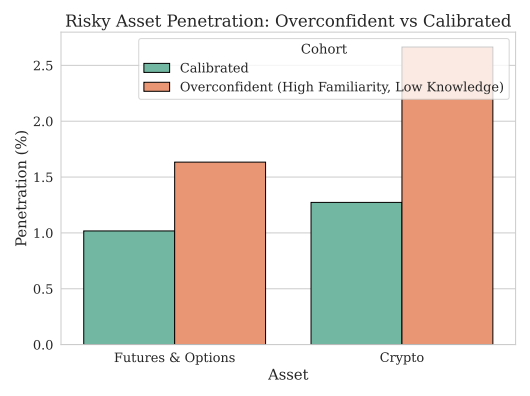

Figure 6: *

Overconfident investors hold risky assets at significantly higher rates than calibrated

Dunning-Kruger effect: Investors with high perceived familiarity but low objective knowledge hold F&O and crypto at significantly higher rates than calibrated peers.

Finfluencer pipeline: 5.0% of investors cite social media as primary information source; this cohort is disproportionately quick-gains oriented relative to professionally-advised investors.

Goal-portfolio gap: The "Daily Trading" cohort has a goal-portfolio consistency score of 7.3%: they identify as speculators without holding the instruments that operationalize that strategy.

Finding 4: Investor Education Programme Impact

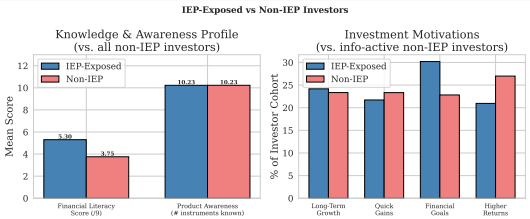

Figure 7: *

IEP-exposed vs. non-IEP investors: knowledge, product awareness, investment motives, and holdings.

2,768 investors (9.9%) cite SEBI or industry IEP as a top information source.

Metric	IEP	Non-IEP
Knowledge score	**5.30**	3.75
Products aware	10.23	10.23
Goal-based (%)	**30.2**	22.8
Quick gains (%)	21.7	23.3
Holds FDs (%)	49.3	**55.1**

The 41% knowledge advantage is specific to *conceptual depth*, not product breadth (awareness is near-identical).

IEP functions as both a knowledge-deepening and a philosophical-reorientation intervention.

Predictive Layer: Three Gradient Boosting Models

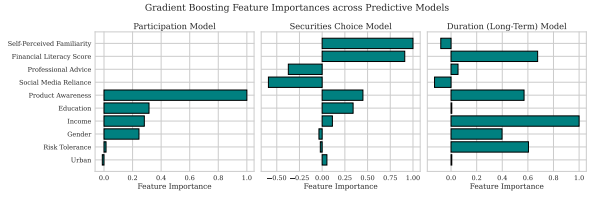

Figure 8: *

Normalized permutation feature importances across all three models.

Model	Task	AUC
M1	Participation	**0.814**
M2	Asset choice	**0.820**
M3	Horizon	0.612

Feature importance shift:
At the entry layer, *product awareness* dominates. Inside the market, *perceived familiarity* and *objective knowledge* take over.

Investment horizon (M3) is the hardest prediction, consistent with it being driven by situational factors outside a cross-sectional feature space.

Interactive Dashboard and Planned Chatbot Layer

Next.js frontend / FastAPI backend

Six interactive sections:

1. **Geographic Penetration** — investor density map across 35 states with per-capita income overlay
2. **Behavioral Patterns** — Dunning-Kruger and finfluencer charts
3. **Household Profile Simulator** — live three-model prediction from user-adjustable inputs
4. **Participation Frontier** — 10×10 income × education probability heatmap
5. **Asset Ladder** — holdings penetration by income quartile
6. **Barrier Wall** — proportional mosaic of

Planned: Chatbot Tab

An additional tab to be added to the existing dashboard that will allow natural language queries over the survey findings, model outputs, and policy recommendations.

Planned architecture:

- LLM with tool-use over the FastAPI endpoints
- Context grounded in the report findings and CSV outputs
- Queries like: "What is the participation probability for a rural woman with 10 years of education?" resolved via live model calls

Policy Recommendations

A. Prioritize product awareness
OR = 2.22; campaigns should name specific instruments (MF, ETF, SIP) for North and South zones.

B. Segment non-investor interventions
Trust-Deficient: regulatory transparency. Knowledge-Gated: vernacular rural delivery. Fear-Driven: first-loss protection onramps.

C. Scale IEP reach
From 9.9% to employer mandates and Jan Dhan partnerships. Content should emphasize conceptual depth, not procedural literacy (KYC awareness is already high). Goal-aligned investors show 72–77% portfolio consistency, confirming goal-based framing works.

D. Close the gender gap
OR = 1.53 for male gender after full controls. SHG-linked investment collectives, women-targeted IEP content, lower minimum investment thresholds.

E. Regulate finfluencers
5% usage associated with quick-gains orientation. Mandatory qualification disclosure, prohibited performance claims, platform liability for unregistered advice.

F. Gate derivatives on objective literacy
Overconfident investors and aspirational daily traders (7.3% consistency) reach leveraged instruments without adequate knowledge. Replace self-certification with a brief objective literacy assessment.

Appendix: Holdings by Income Quartile

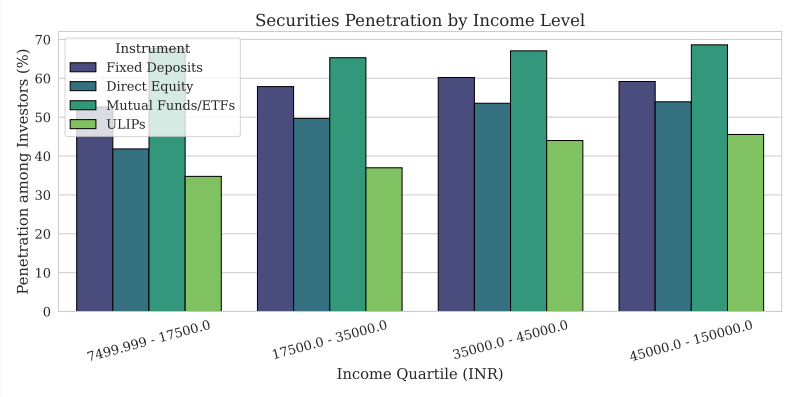

Figure 9: *

Market-linked instruments scale sharply with wealth.

Appendix: Goal-Portfolio Consistency

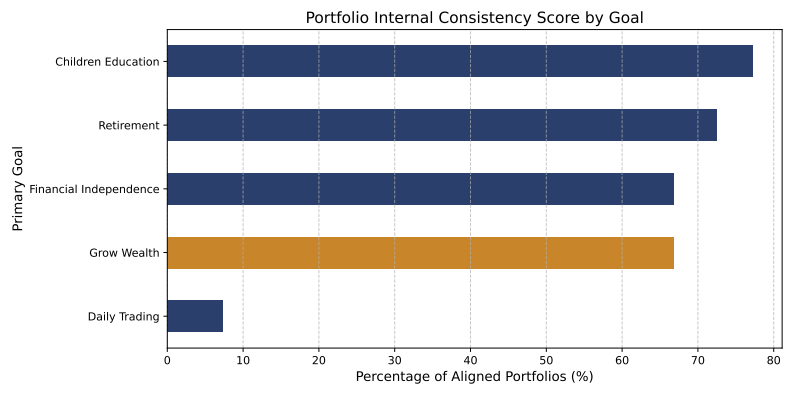

Figure 10: *

Consistency score by stated primary goal. Daily Trading cohort: 7.3%.

Appendix: Investment Horizons by Asset Class

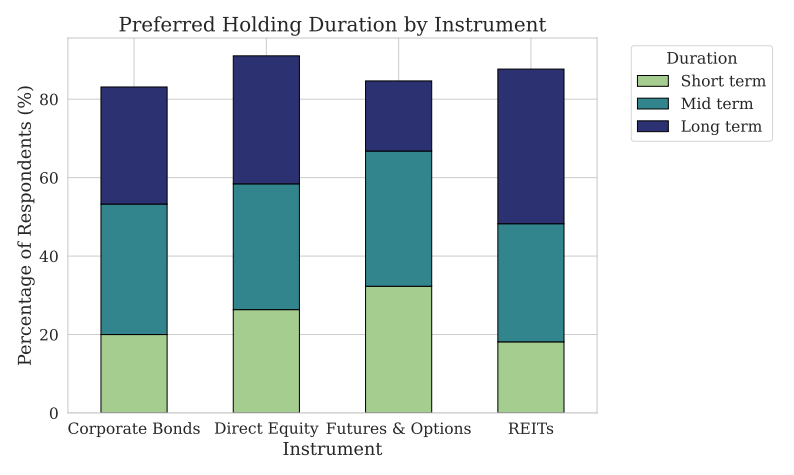

Figure 11: *

Appendix: Full Logistic Regression Table

Predictor	**OR**	**95% CI**	z	Sig.
Product Awareness	2.218	[2.166, 2.272]	65.2	***
Log(Income)	1.603	[1.564, 1.643]	37.4	***
Education (yrs)	1.556	[1.518, 1.594]	35.6	***
Male	1.526	[1.490, 1.564]	33.9	***
Urban	1.455	[1.422, 1.489]	32.0	***
Risk Tolerance	1.170	[1.145, 1.196]	14.0	***
Millennial	1.124	[1.098, 1.151]	9.8	***
Gen X	0.993	[0.969, 1.017]	−0.6	
Baby Boomer	0.907	[0.881, 0.934]	−6.6	***
North	0.850	[0.825, 0.876]	−10.7	***
South	0.842	[0.816, 0.869]	−10.7	***
West	1.021	[0.989, 1.054]	1.3	

$N = 97{,}626$ McFadden $R^2 = 0.290$ *** $p < 0.001$, HC3 robust SE